

Image Analysis Report

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Executive Summary

The image exhibits high severity localized corrosion, primarily concentrated along a horizontal seam with evidence of deep pitting and significant material loss. The extent of coating failure and heavy scaling suggests a high risk of through-wall penetration in the central defect area.

Key Findings

#	Finding
1	Severe localized corrosion with heavy scaling and rust buildup centered on a horizontal weld or structural joint.
2	Widespread coating failure characterized by delamination, flaking, and blistering surrounding the primary defect.
3	Evidence of deep pitting under the concentrated iron oxide deposit in the central region.
4	Extensive rust staining and run-off indicating active atmospheric corrosion and moisture entrapment at the seam.
5	Substrate material loss is visually apparent, suggesting significant wall thinning of the base metal.

Localized detections (VLM, approximate)

- Severe Localized Corrosion (conf. 95%): x=0.340, y=0.120, w=0.450, h=0.780
- Seam Corrosion (conf. 88%): x=0.000, y=0.440, w=1.000, h=0.120

Recommendations

#	Recommendation
1	Conduct immediate Ultrasonic Thickness (UT) testing at the central defect and along the horizontal seam to determine the remaining wall thickness.
2	Perform abrasive blast cleaning to SSPC-SP 10 (Near-White Blast Cleaning) to remove all corrosion products and allow for accurate pit depth measurement.
3	Evaluate the need for structural repair (e.g., welded patch or composite wrap) if wall thickness is found to be below the minimum allowable (MAWT) per ASME PCC-2.
4	Apply a high-performance protective coating system including a zinc-rich primer to the properly prepared surface.
5	Assess the horizontal seam for potential crevice corrosion or improper seal that may be trapping corrosive agents.
6	Increase the inspection frequency for this asset to monitor for any recurrence of localized high-rate corrosion.